SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS **COURSE CODE: MLA01**

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 40 %

Total Marks: 50 **Time: 60 Mins**
No. of Questions: 5

Annexure A

Agent Design Assessment

Code of Conduct:

I SHAM S. – 192325076 _ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Project Overview

Design and develop a **Smart Student Support AI Agent** that provides students with instant, intelligent, and personalized assistance regarding academic and campus-related services. The AI agent should simulate a virtual university assistant capable of understanding natural language, guiding students through different services, maintaining conversational context, and delivering accurate responses in real time.

The interface should be modern, clean, responsive, and similar to ChatGPT, with a university-inspired color theme. The AI agent should provide a professional user experience suitable for students, faculty, and administrative staff.

WHAT is the project?

The project is an **AI-powered virtual assistant** developed to help university students quickly access information about academic and campus services through natural conversation.

Unlike a traditional FAQ page, the AI agent should interact conversationally, understand different ways users ask the same question, remember the current conversation when appropriate, and guide users toward the information they need.

The assistant should support both menu-driven navigation and free-text queries.

WHY is this project needed?

University students often spend time searching different websites or visiting administrative offices for routine information.

The AI agent aims to:

- Provide instant support 24/7.
- Reduce waiting time for common inquiries.
- Improve the student experience through conversational interaction.

- Minimize repetitive administrative work.
- Serve as a centralized source for academic and campus information.
- Demonstrate the practical use of AI agents in educational environments.

WHO will use the AI Agent?

Primary Users

- Undergraduate students
- Postgraduate students
- New admissions

Secondary Users

- Faculty members
- Administrative staff
- Visitors seeking campus information

WHERE will the AI Agent be used?

The AI agent should be accessible from:

- University website
- Student portal
- Department portal
- Mobile devices
- Desktop browsers
- Campus kiosks

WHEN will students use it?

Students should be able to interact with the AI agent:

- During admissions
- Course registration periods
- Before examinations
- During semester classes
- While checking attendance
- For campus navigation
- Anytime throughout the day

The assistant should be available **24×7**.

HOW does the AI Agent work?

The AI agent should follow this workflow:

```
Student Opens AI Agent

↓

Welcome Screen

↓

Student Types Question
OR
Selects Quick Action

↓

AI Agent Understands Intent

↓

Analyzes Context

↓

Retrieves Appropriate Information

↓

Displays Response

↓

Suggests Related Questions

↓

Student Continues Conversation

↓

Conversation Ends
```

WHAT should the AI Agent do?

The AI agent should assist students with:

Academic Services

- Course Registration
- Timetable
- Examination Schedule
- Attendance Information
- Results
- Academic Calendar

Campus Services

- Library
- Hostel
- Transport
- Cafeteria
- Wi-Fi Support
- Campus Map

Student Services

- Scholarships
- Placement Cell
- Student Clubs
- Faculty Contacts
- Fee Payment Information

General Support

- Frequently Asked Questions
- Contact Office
- Feedback Collection

HOW should conversations work?

Example 1

Student

I want to register for next semester.

AI Agent

I can help with course registration.
Would you like to know the registration dates, required documents, or the registration process?

Example 2

Student

When is my AI exam?

AI Agent

Please tell me your department and semester so I can provide the correct examination schedule.

Example 3

Student

Where is the library?

AI Agent

The Central Library is located near Block B.
It is open from 8:00 AM to 8:00 PM.

HOW should the User Interface look?

Home Screen

Display:

- University Logo
- AI Agent Avatar
- Welcome Message
- Search Bar
- Start Chat Button

Chat Interface

The chat window should resemble modern AI assistants.

Include:

- Chat bubbles
- Typing animation
- AI avatar
- User avatar
- Timestamp
- Suggested questions
- Scrollable conversation

Quick Action Cards

Instead of plain buttons, use interactive cards.

Examples:

🗐 Course Registration

📅 Timetable

Examination

Attendance

Campus Services

Campus Map

Contact Office

Sidebar

Include:

Dashboard

Recent Conversations

Student Services

Notifications

Settings

Help

About

Logout

Dashboard Statistics

Show cards displaying:

Students Assisted Today

Active Conversations

Average Response Time

Frequently Asked Questions

Student Satisfaction

AI Features

The AI agent should support:

- Natural language understanding
- Context-aware conversations
- Multi-turn dialogue
- Personalized responses
- Suggested follow-up questions
- Smart navigation
- Intelligent recommendations

HOW should unknown questions be handled?

If the AI agent cannot answer:

Display

I'm sorry, I couldn't find an exact answer for your question.

Offer:

- Related suggestions
- Contact Office
- Return to Main Menu

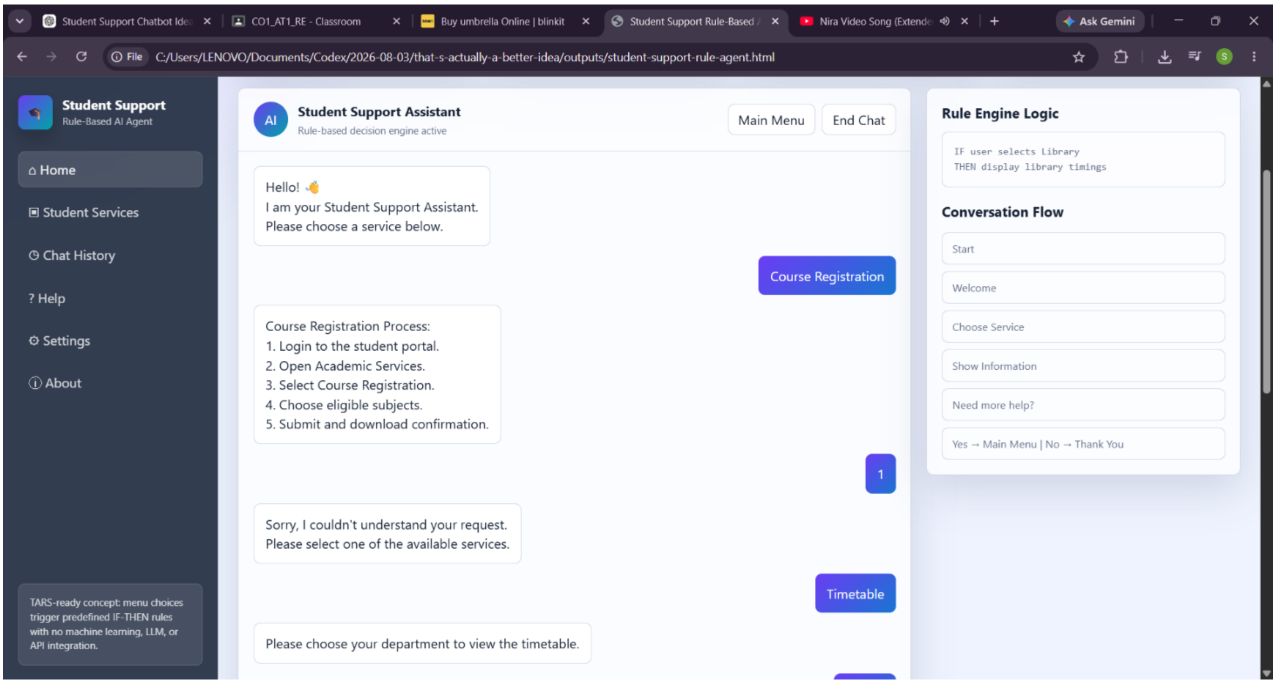
HOW should navigation work?

```
Home

↓

Start Chat

↓

User Question

↓

AI Understanding

↓

Relevant Service

↓

Provide Answer

↓

Related Suggestions

↓

Continue Chat

↓

End Conversation
```

UI Design Requirements

The interface should be:

- Modern
- Responsive
- Professional
- Mobile Friendly
- University Theme
- Purple and Blue Color Palette
- Rounded Components
- Glassmorphism Effects
- Soft Shadows
- Smooth Animations
- Dark Mode
- ChatGPT-inspired Layout



Expected Features

- AI-powered conversation
- Smart question understanding
- Context-aware dialogue
- Interactive dashboard
- Quick actions
- Search functionality
- Suggested follow-up questions
- Conversation history
- Responsive design
- Professional UI

Q. No.	Criteria to be evaluated	BL	Marks
1	**Problem Analysis and Agent Design:** Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	**Dialogflow Agent Development:** Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10

3	**Problem Formulation and Decision Flow:** Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	**Search Strategy Application:** Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	**Testing and Evaluation:** Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.

Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.